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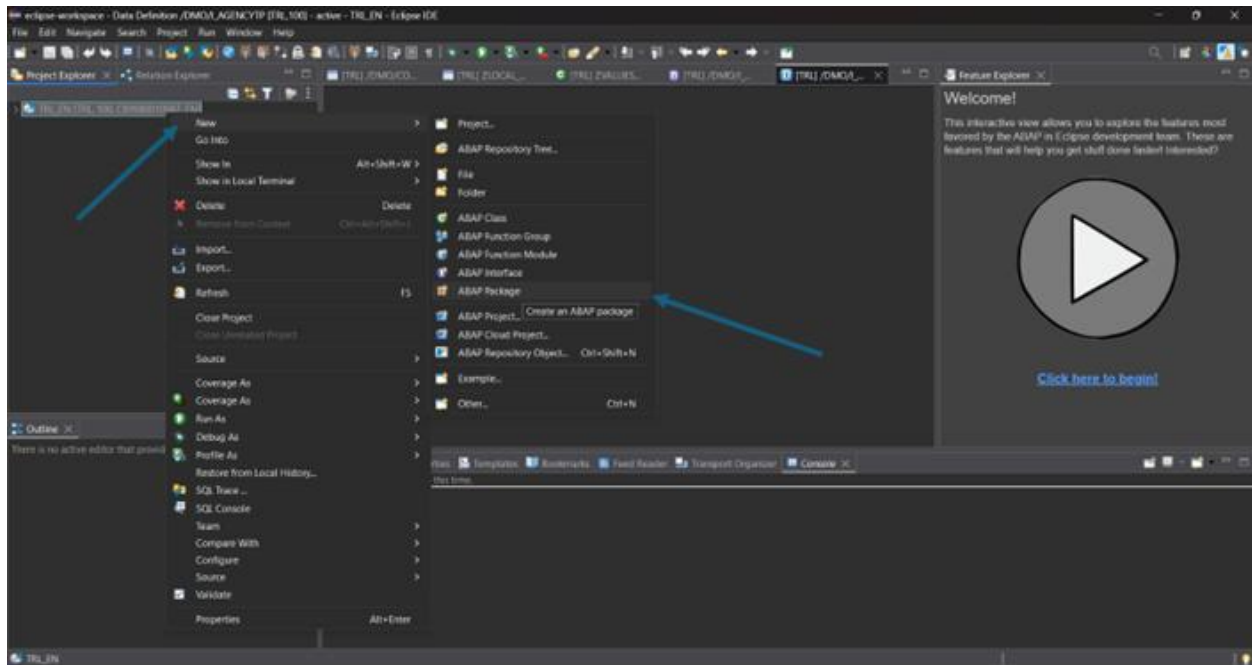
LAB MANUAL

ABAP RESTful Application Programming Model

Building ABAP Restful Programming:

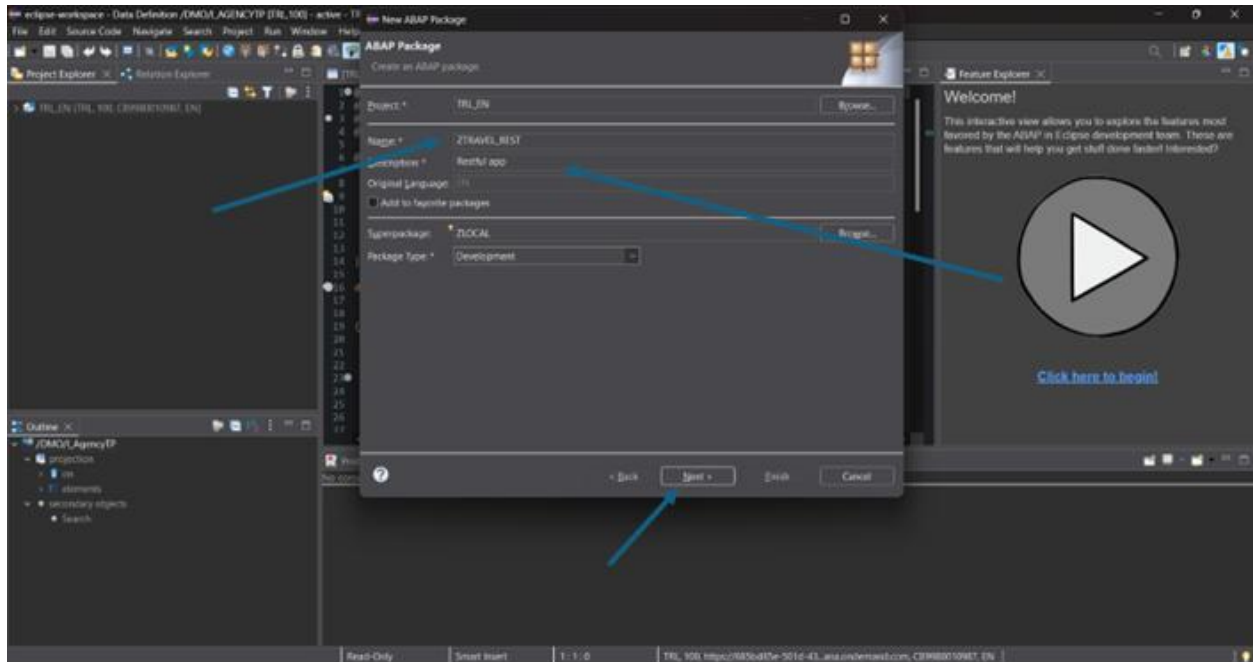
Step 1:- Creating the new package

- Right click on the “TRL_EN”
- Click on “New” and Click on ABAP Package.



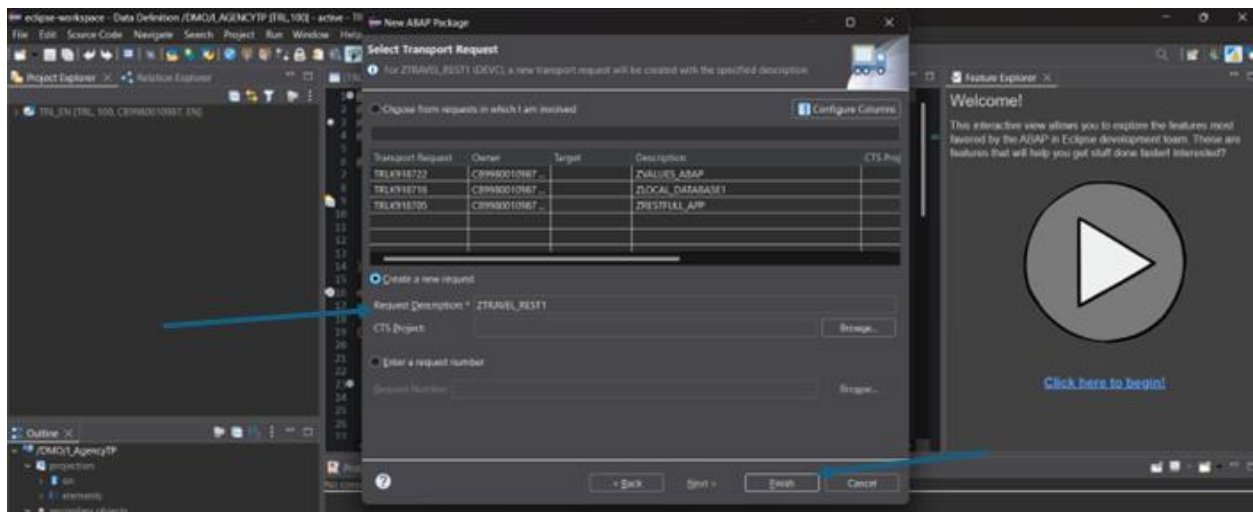
Step 2:- Creating a new ABAP Package following with the name and description

- Name : ZTRAVEL_REST_11
- Description: Restful app
- Superpackage: ZLOCAL
- Click on “Next” and Click on “Next”



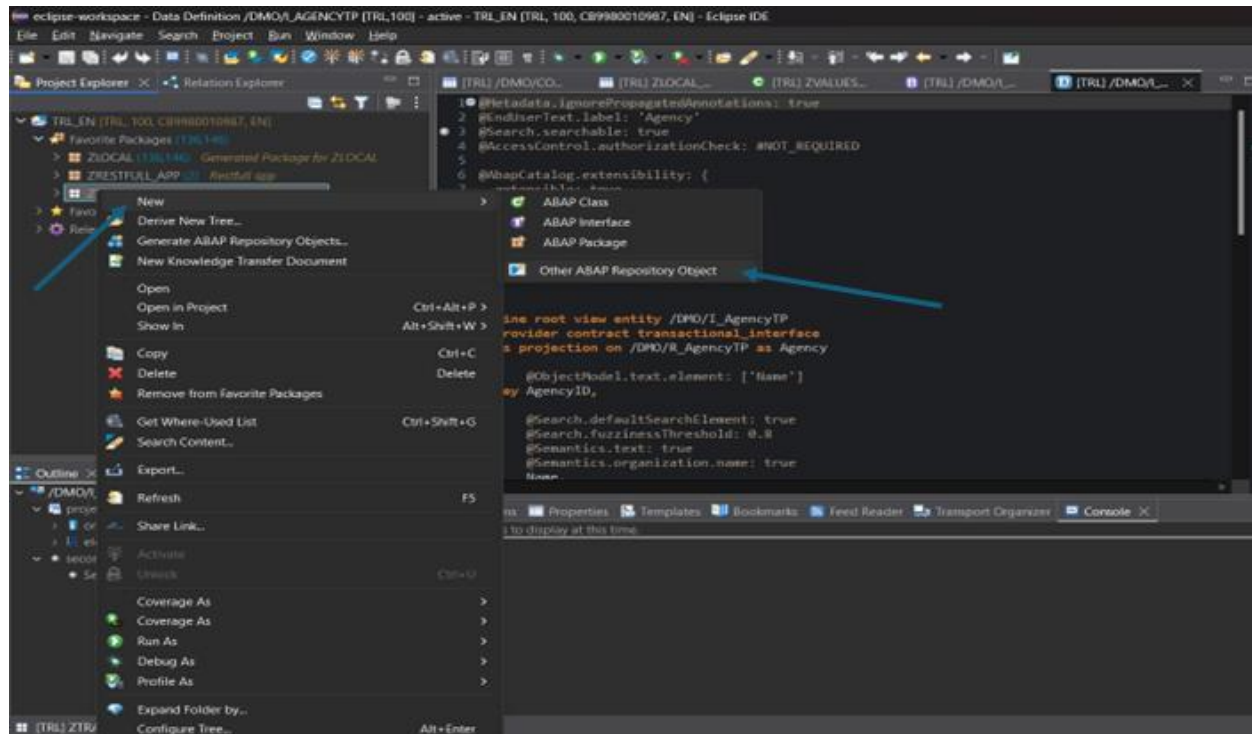
Step 3:- Click on the Create new request

- Request Description : ZTRAVEL_REST1
- Click on "Finish"



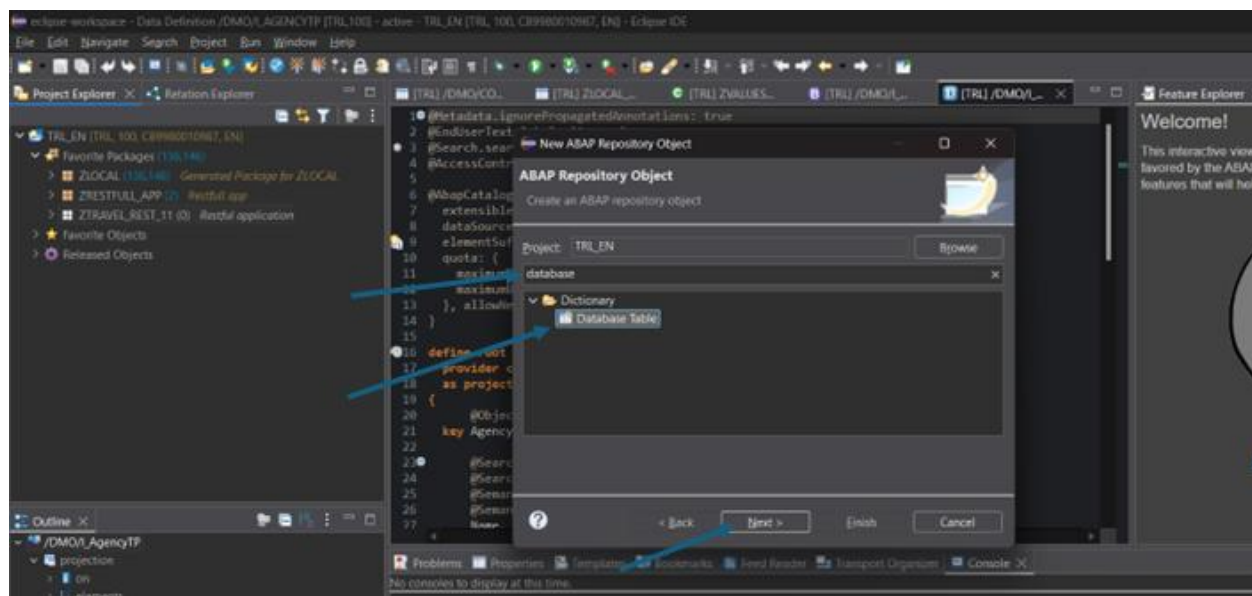
Step 4:- Creating a Database table

- Right click on the ZTRAVEL_REST_11
- Click on “New” and Click on “Other ABAP Repository Object”



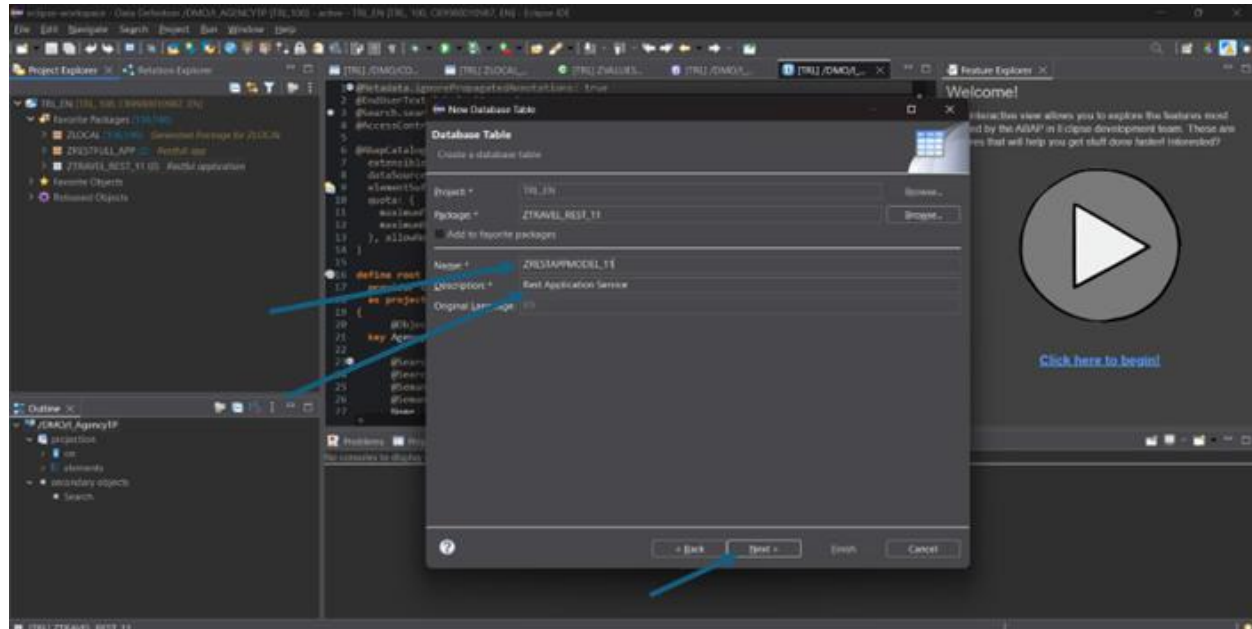
Step 5:- Search for the Database table

- Click on the Database table
- Click on “Next”



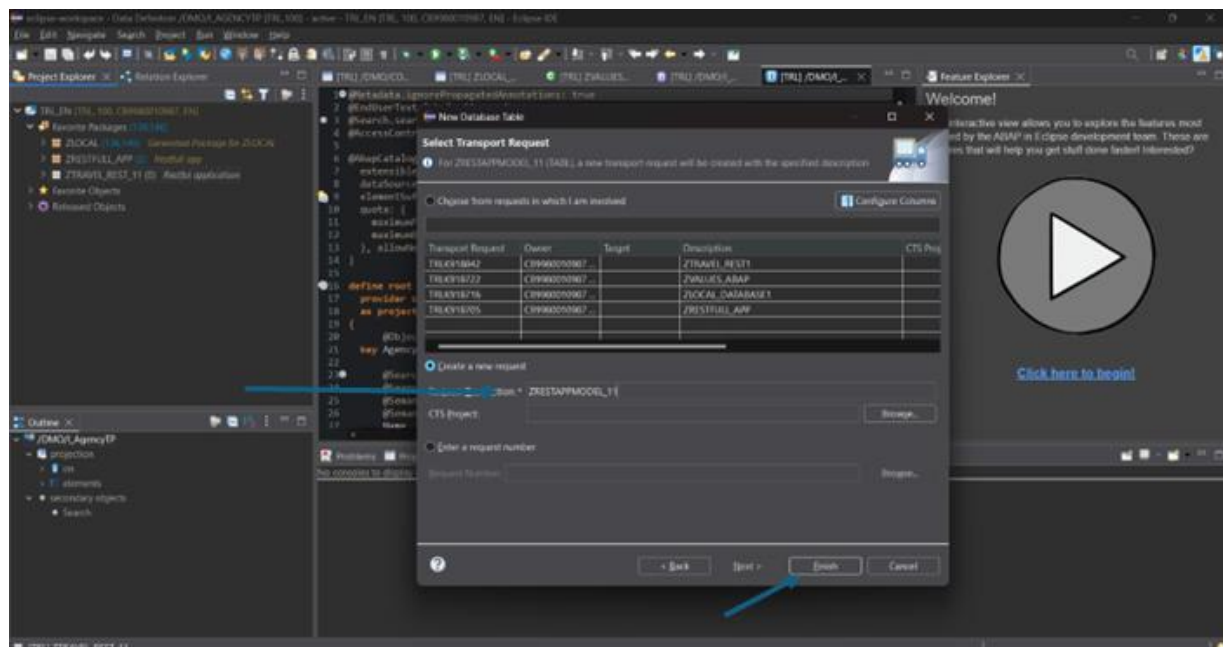
Step 6:- Creating a new Database table following with the name and description.

- Name: ZRESTAPPMODEL_11
- Description: Restful application Service
- Click on “Next”.

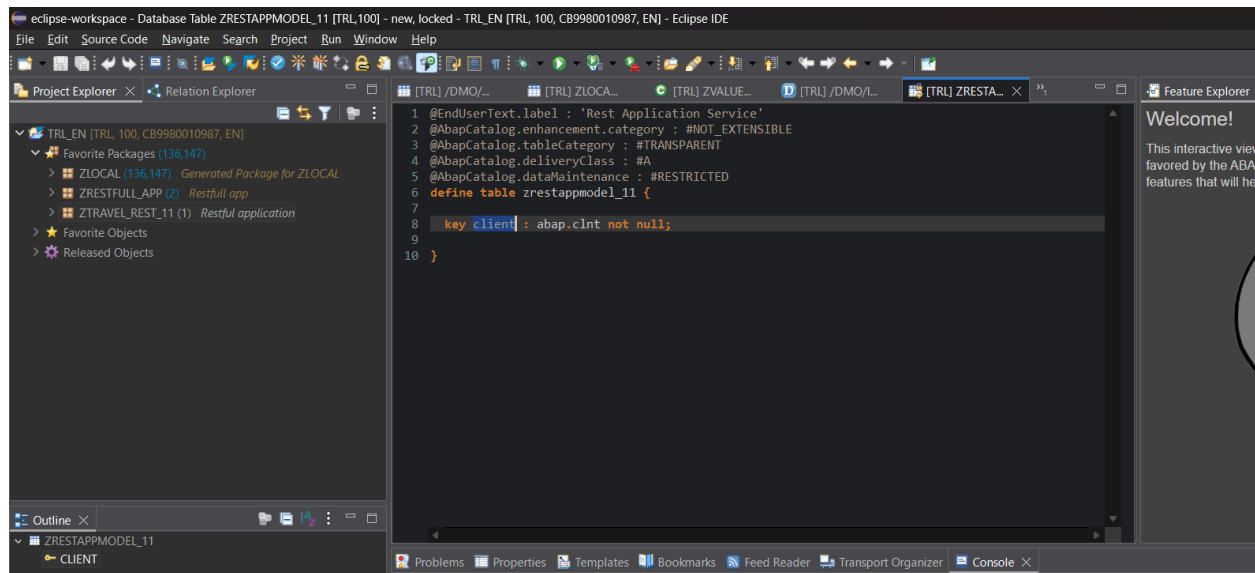


Step 7:- Click on the Create new request

- Request Description : ZRESTAPPMODEL_11
- Click on “Finish”



Step 8:- Ones you click on Finish the database table is created



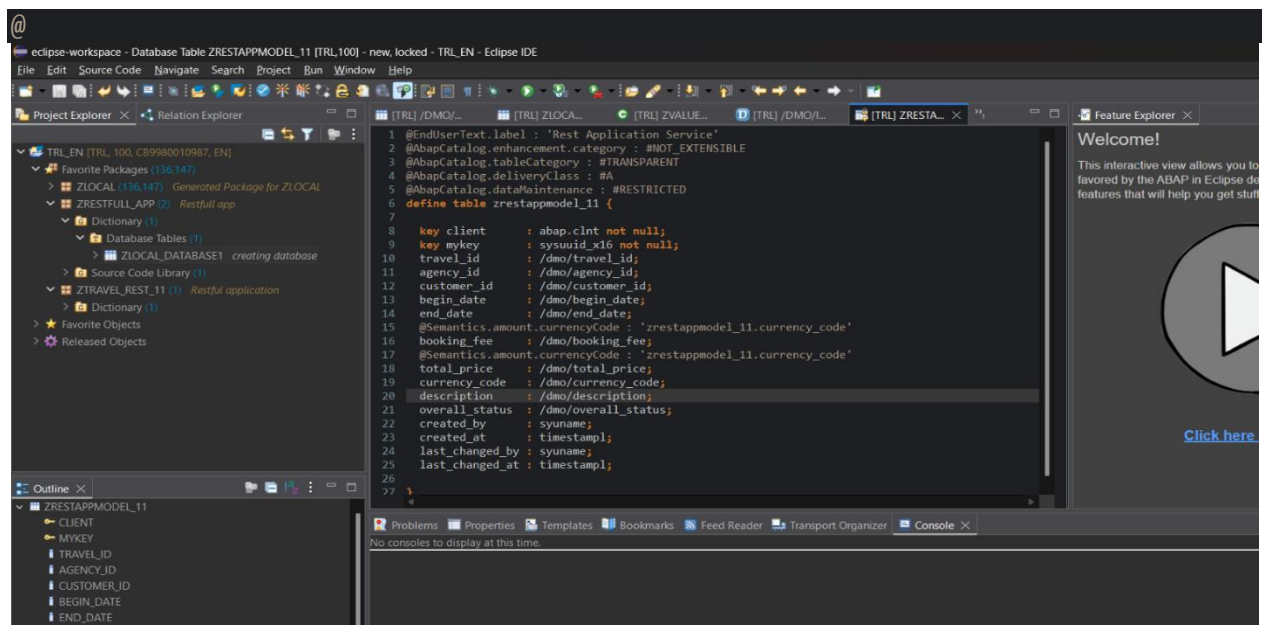
Step 9:- Insert the give code for the creating table

```

@EndUserText.label : 'Rest Application Service'
@AbapCatalog.enhancement.category : #NOT_EXTENSIBLE
@AbapCatalog.tableCategory : #TRANSPARENT
@AbapCatalog.deliveryClass : #A
@AbapCatalog.dataMaintenance : #RESTRICTED
define table zrestappmodel_11 {

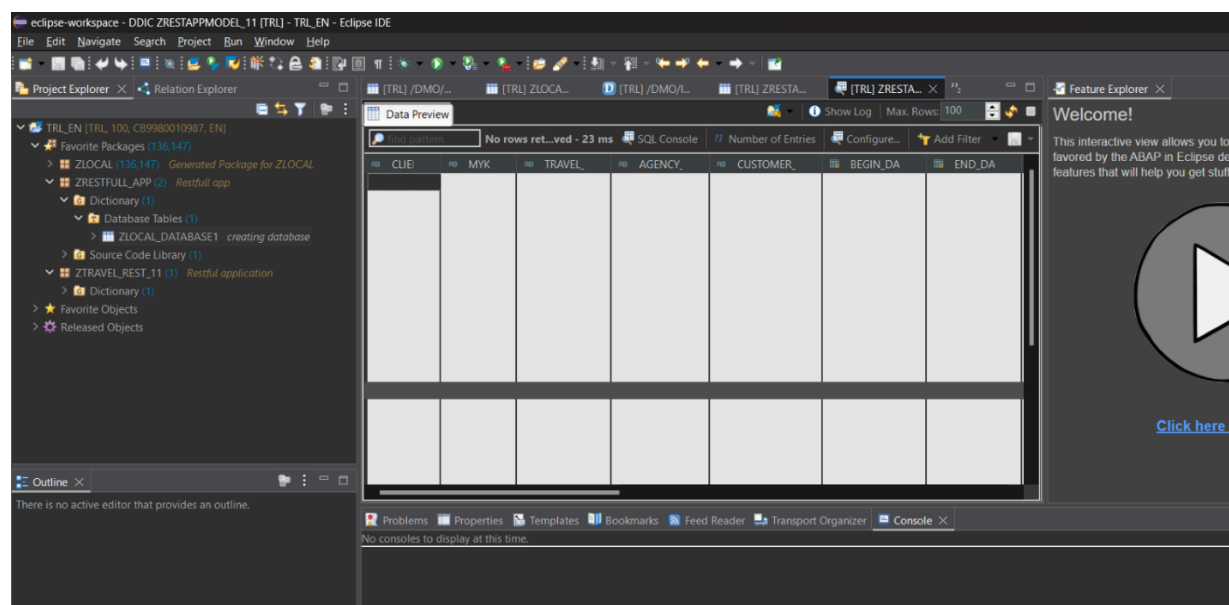
key client : abap.clnt not null;
key travel_id : /dmo/travel_id not null;
agency_id : /dmo/agency_id;
customer_id : /dmo/customer_id;
begin_date : /dmo/begin_date;
end_date : /dmo/end_date;
@Semantics.amount.currencyCode : 'zrestappmodel_11.currency_code'
booking_fee : /dmo/booking_fee;
@Semantics.amount.currencyCode : 'zrestappmodel_11.currency_code'
total_price : /dmo/total_price;
currency_code : /dmo/currency_code;
description : /dmo/description;
overall_status : /dmo/overall_status;
attachment : /dmo/attachment;
mime_type : /dmo/mime_type;
file_name : /dmo/filename;
created_by : abp_creation_user;
created_at : abp_creation_tstmpl;
local_last_changed_by : abp_locinst_lastchange_user;
last_changed_at : abp_lastchange_tstmpl;
local_last_changed_at : abp_locinst_lastchange_tstmpl;

```



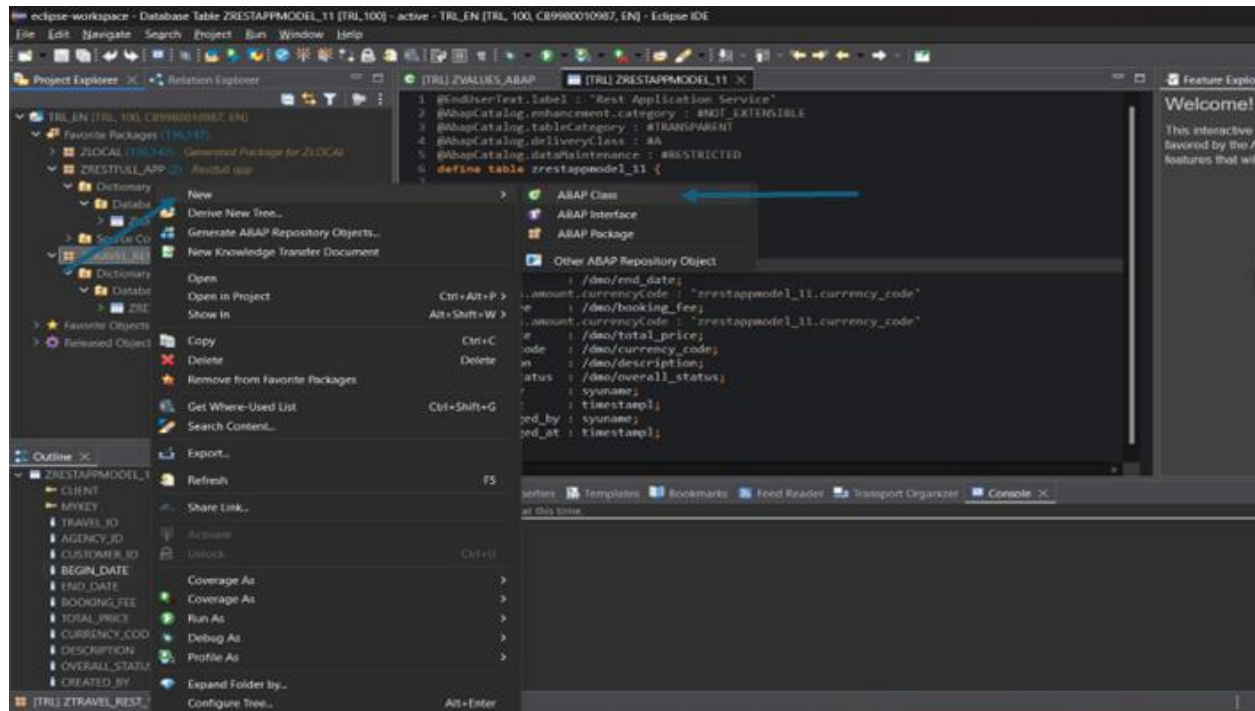
Step 10:- Follow the following instructions and Run the code.

- CTRL+A:-Select whole code
- CTRL+S:- Save the code
- CTRL+F2:- To check the syntax of code
- CTRL+F3:- To activate the script
- Press F8 to check the preview of table.



Step 11:- Create a ABAP class for inserting the values in database.

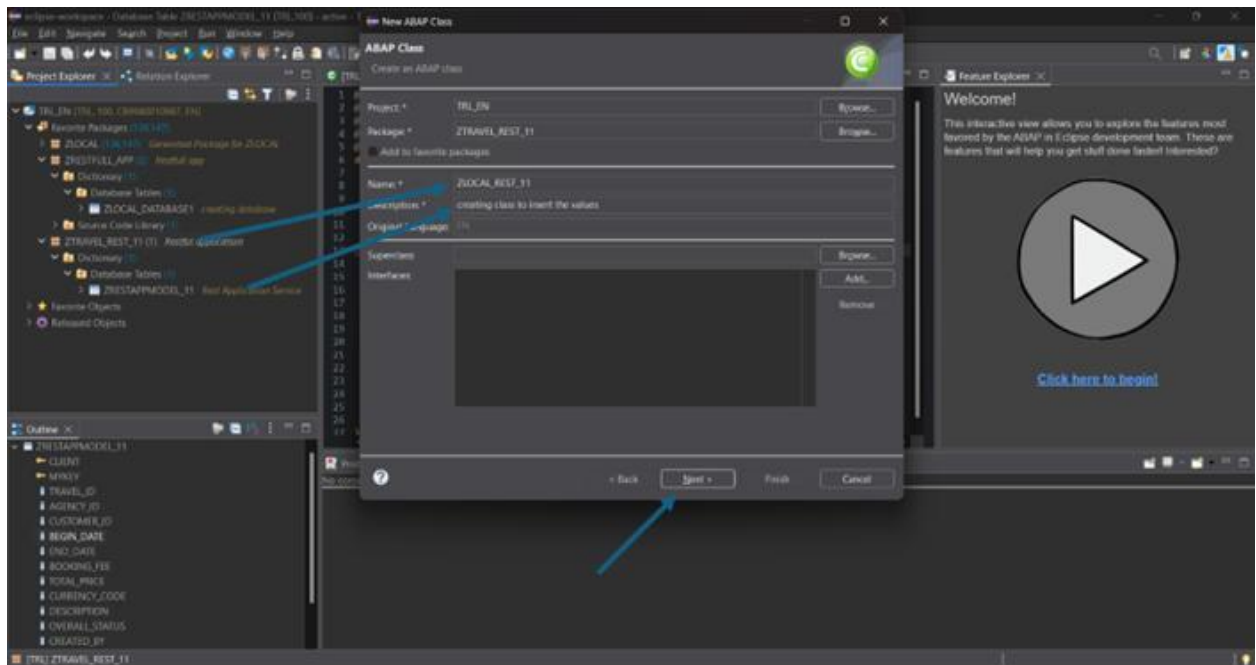
- Right click on the “ZTRAVEL_REST_11”
- Click on “New”
- Click on “ABAP Class”



Step 12:- Creating a new class table following with the name and description.

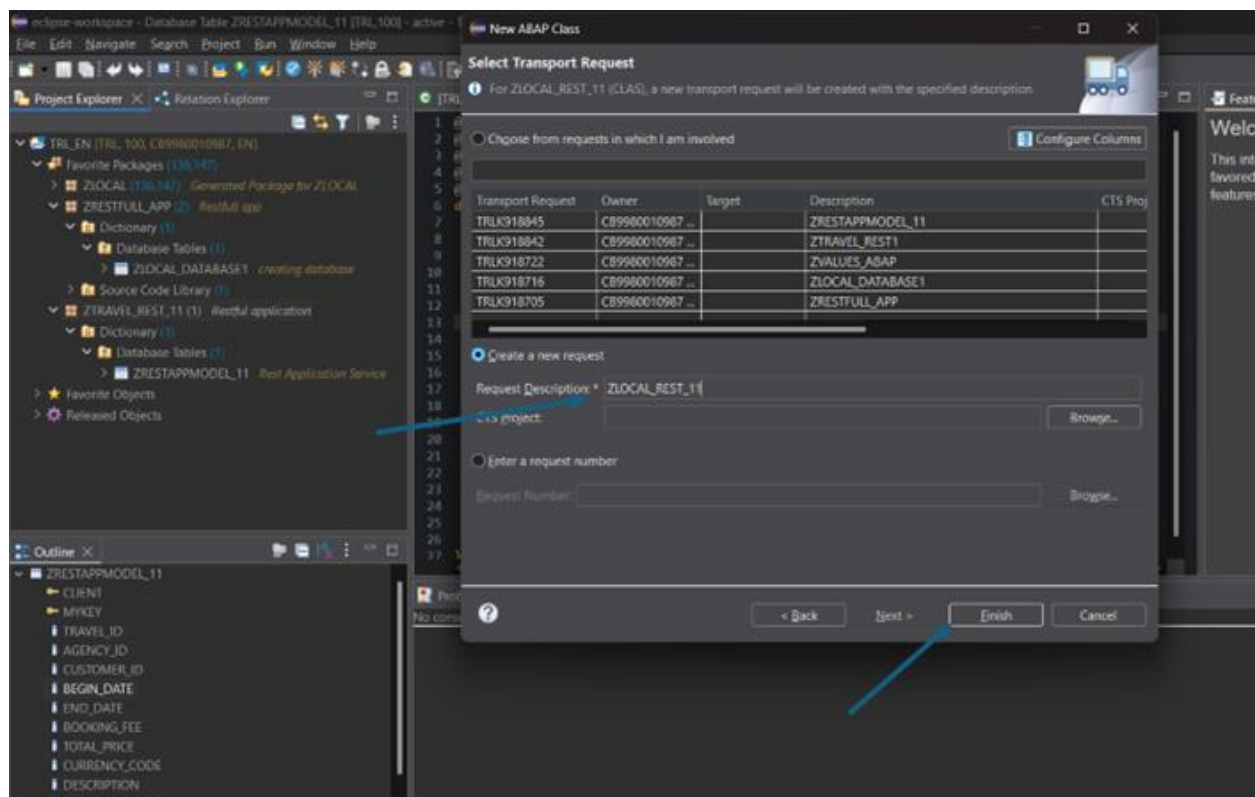
- Name :- ZLOCAL_REST_11.
- Description:- inserting the values.

- Click on “Next”



Step 13:- Click on the “Create a new request”

- Request description: ZVALUES_ABAP
- Click on “Finish”



Step 14:- Inserting the values into Database table

```
CLASS zlocal_rest_11 DEFINITION
PUBLIC
FINAL
CREATE PUBLIC .

PUBLIC SECTION.
INTERFACES if_oo_adt_classrun .
PROTECTED SECTION.
PRIVATE SECTION.
ENDCLASS.

CLASS zlocal_rest_11 IMPLEMENTATION.
METHOD if_oo_adt_classrun~main.

"data decleration
DATA:
attachment TYPE /dmo/attachment,
file_name TYPE /dmo/filename,
mime_type TYPE /dmo/mime_type.

"clear travel table
DELETE FROM zrestappmodel_11.

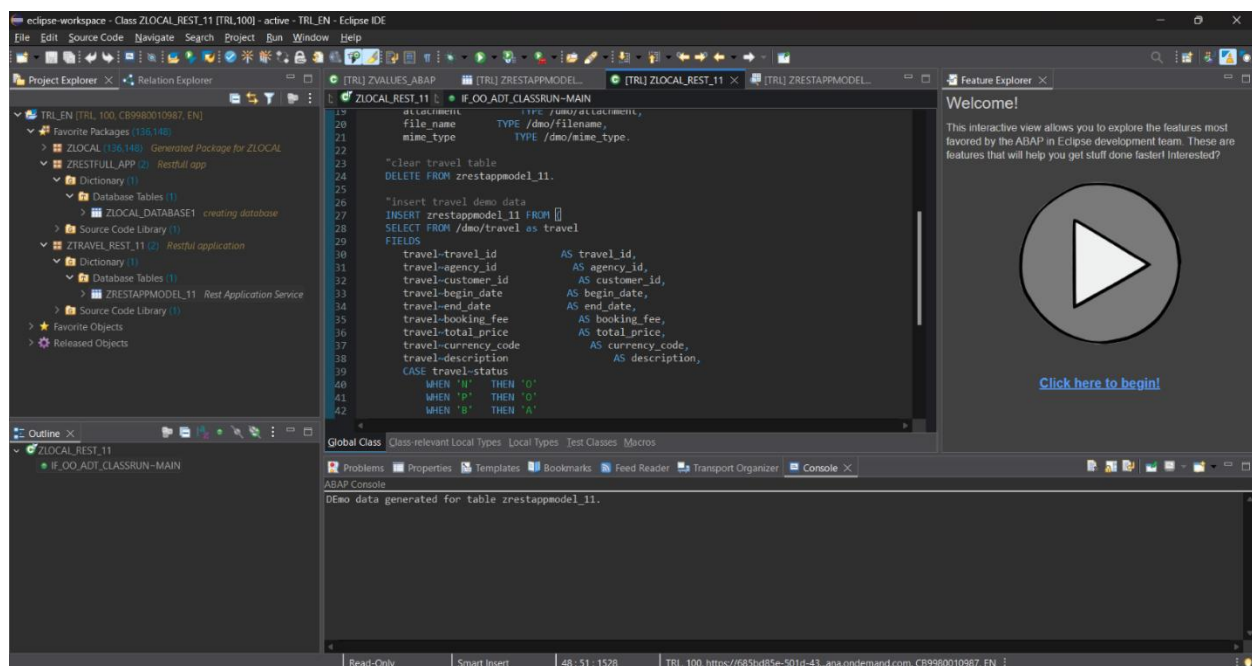
"insert travel demo data
INSERT zrestappmodel_11 FROM (
SELECT FROM /dmo/travel as travel
FIELDS
travel~travel_id AS travel_id,
travel~agency_id AS agency_id,
travel~customer_id AS customer_id,
travel~begin_date AS begin_date,
travel~end_date AS end_date,
travel~booking_fee AS booking_fee,
travel~total_price AS total_price,
travel~currency_code AS currency_code,
travel~description AS description,
CASE travel~status
WHEN 'N' THEN 'O'
WHEN 'P' THEN 'O'
WHEN 'B' THEN 'A'
ELSE 'X'
END AS overall_status,
@attachment AS attachment,
@mime_type AS mime_type,
@file_name AS file_name,
```

```

travel~createdby AS created_by,
travel~createdat AS created_at,
travel~lastchangedby AS last_changed_by,
travel~lastchangedat AS last_changed_at,
travel~lastchangedat AS local_last_changed_at
ORDER BY travel_id UP TO 10 ROWS

).
COMMIT WORK.
out->write( |DEmo data generated for table zrestappmodel_11. | ).
ENDMETHOD.
ENDCLASS.

```

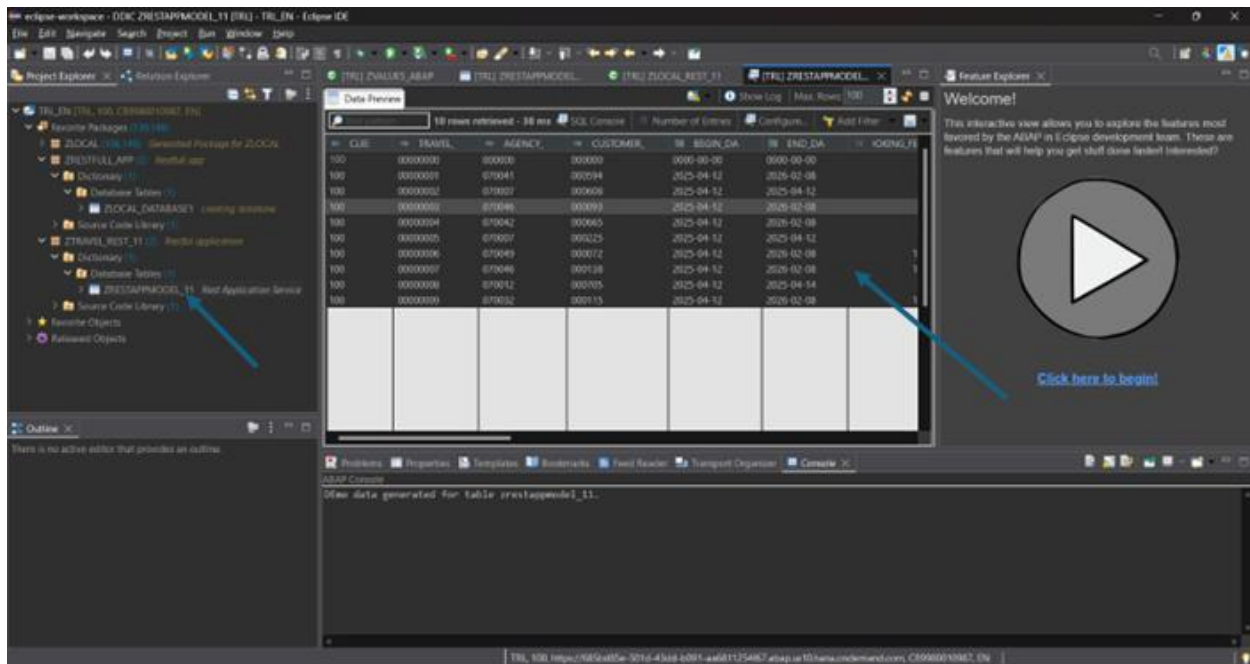


Step 15:- Follow the following instructions and Run the code.

- CTRL+A:-Select whole code
- CTRL+S:- Save the code
- CTRL+F2:- To check the syntax of code
- CTRL+F3:- To activate the script

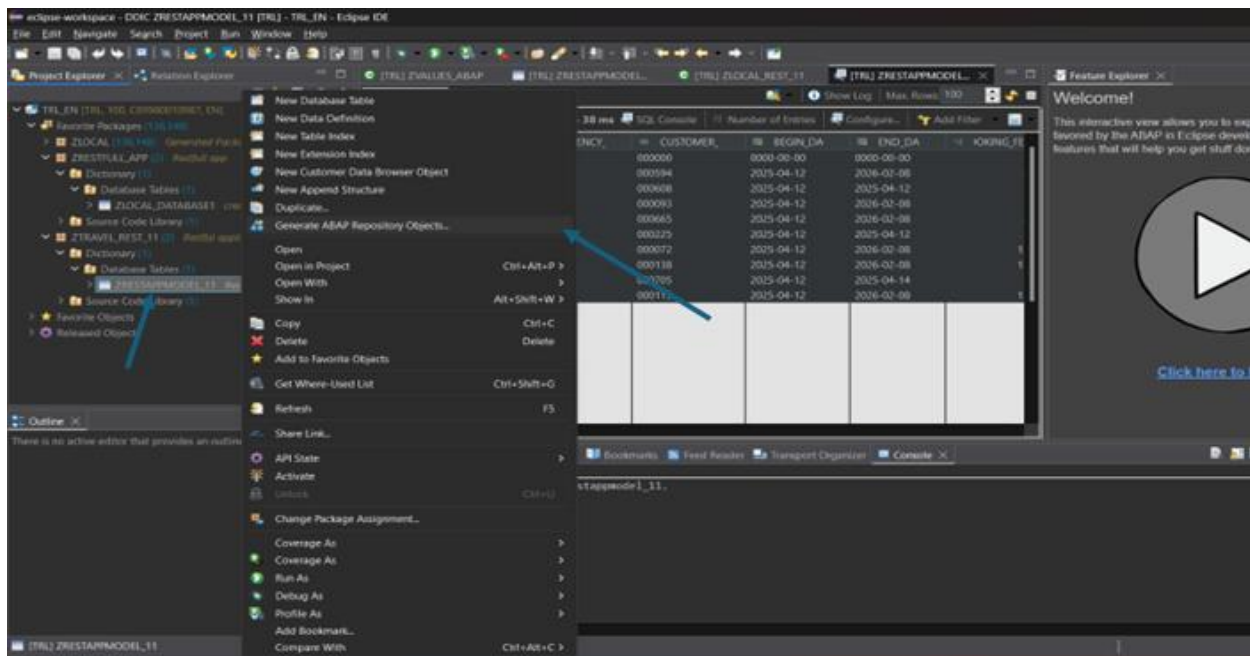
Step 16:- Click on the values inserted or not

- Click on the database you created “zrestappmodel_11”
- Press F8 to check the values inserted or not.

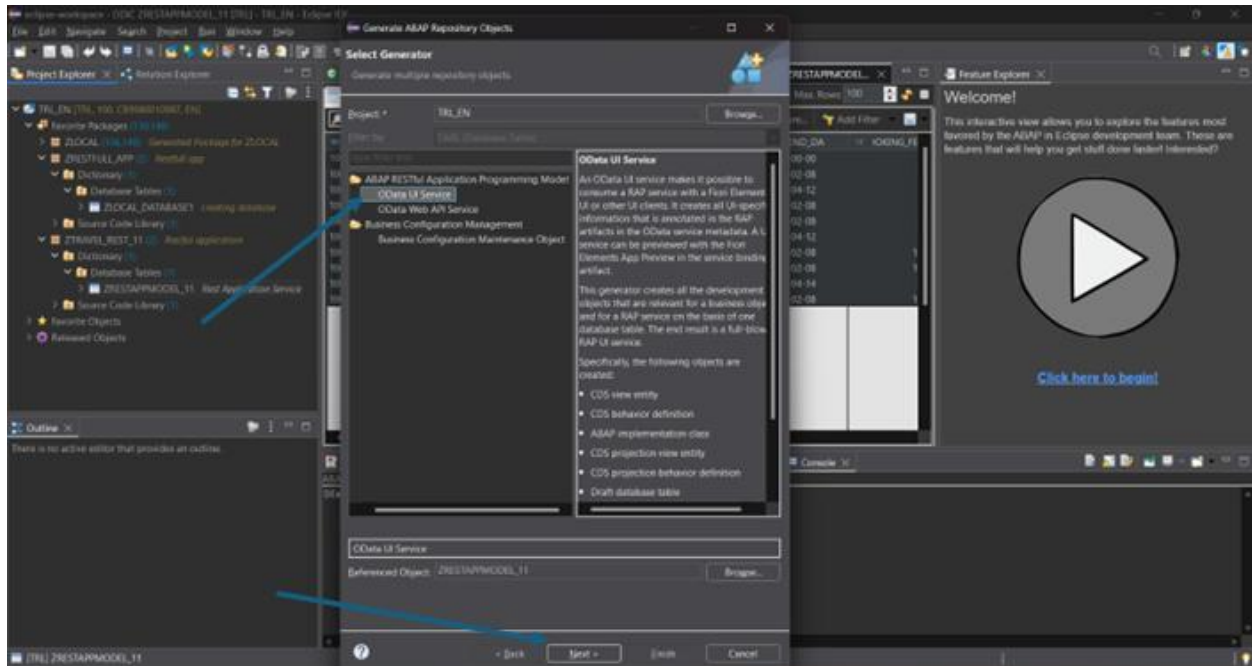


Step 17:- Creating RESTFUL Application

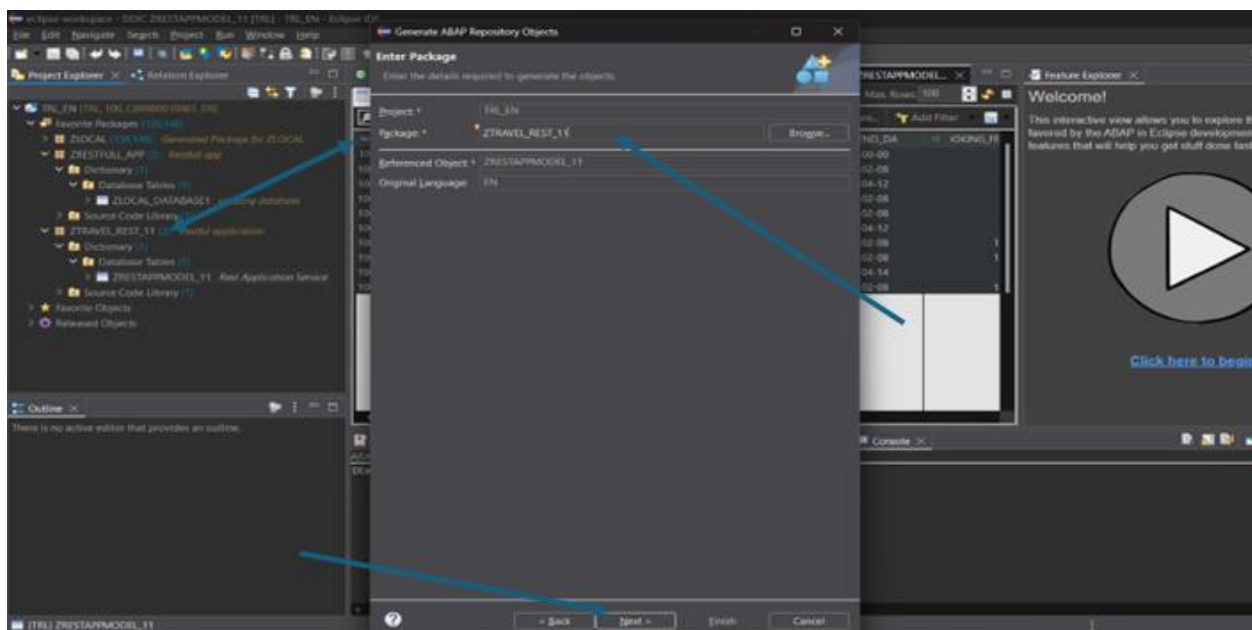
- Right click on the database table.
- Click on the “Generate ABAP Repository Object”



Step 18:- Click on the “OData UI Service” and Click on “Next”

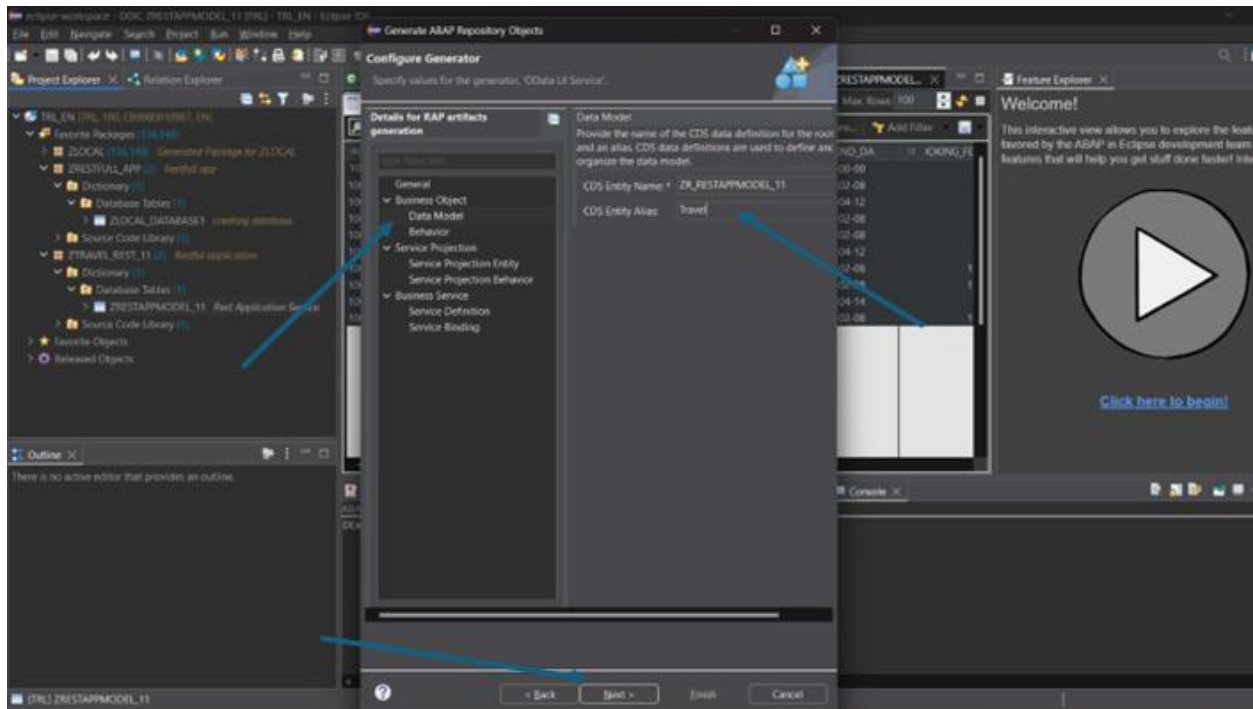


Step 19:- Enter the Your Package Name “ZTRAVEL_REST_11” and Click on “Next”

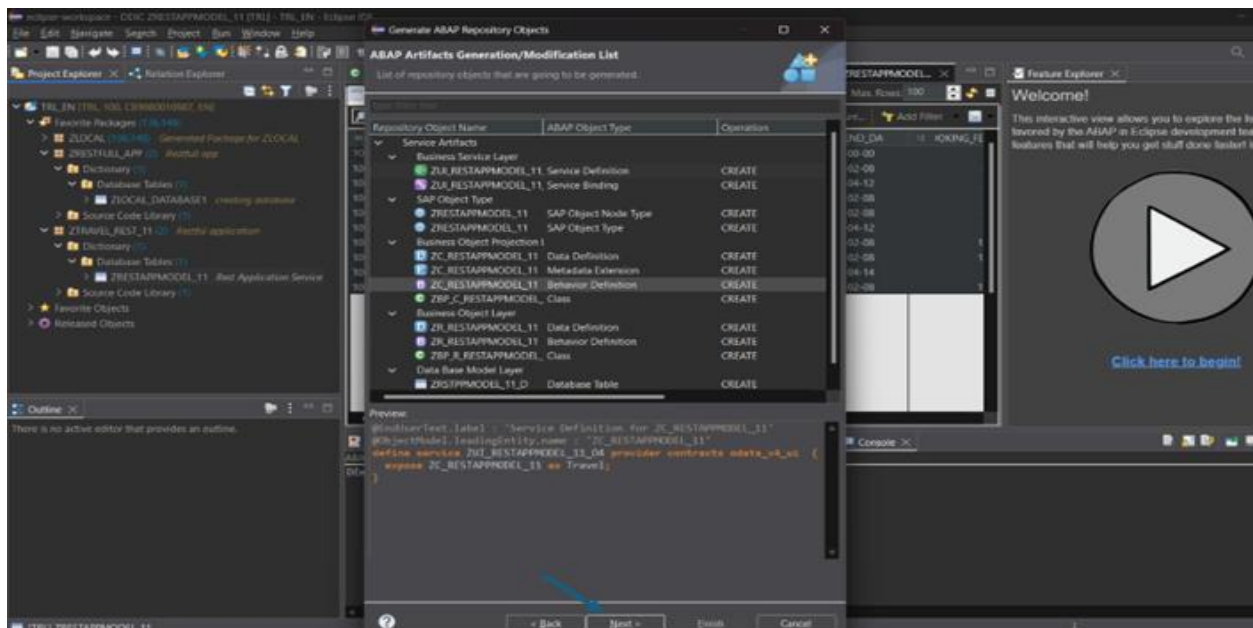


Step 20:-Click on the “Data Model”

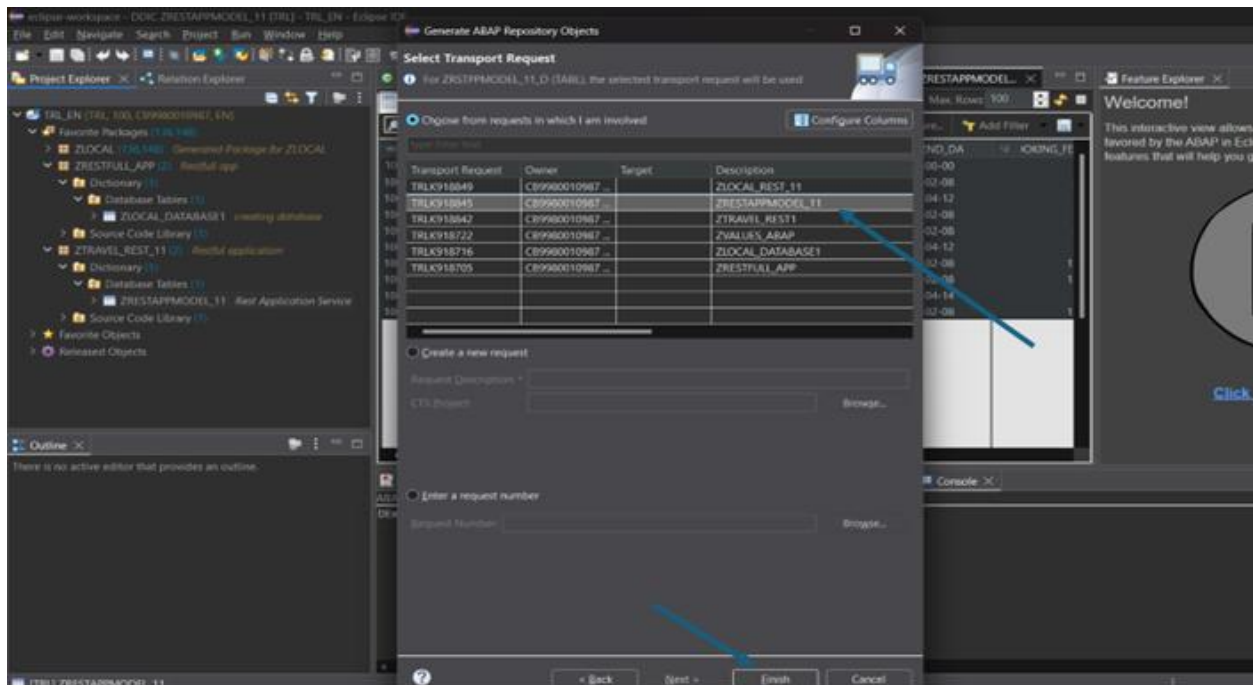
- CDS Entity Alias : “You can give any name on your wish”
- Click on ”Next”



Step 21:- Click on “Next”

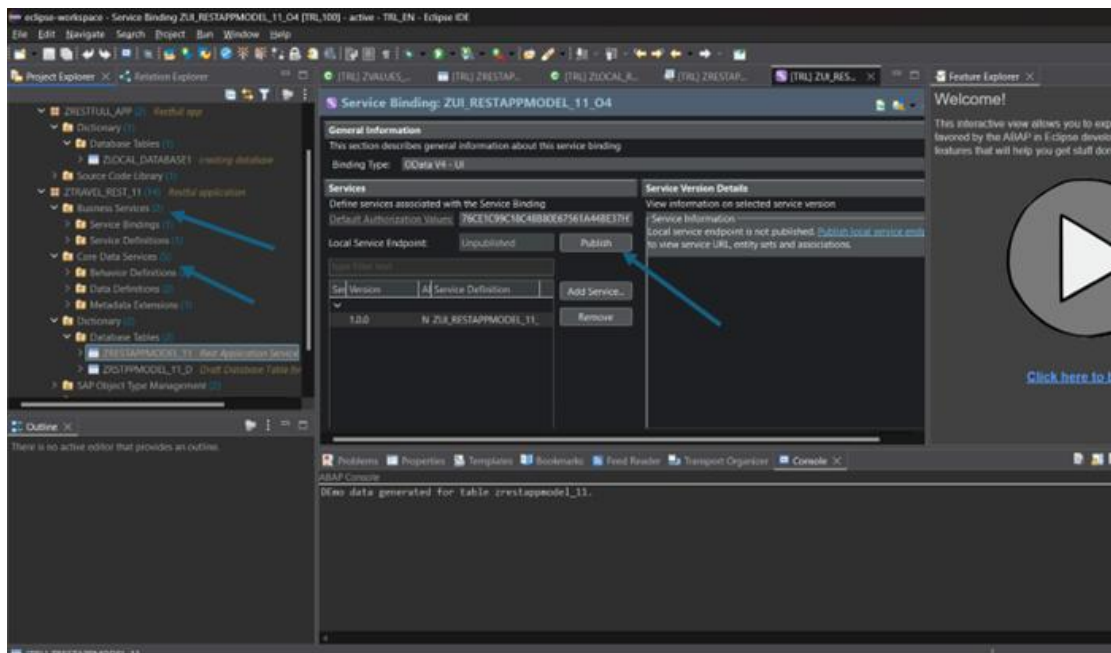


Step 22:- Click on the “ZRESTAPPMODEL_11” and Click on “Finish”



Step 23:- Once you click on finish,

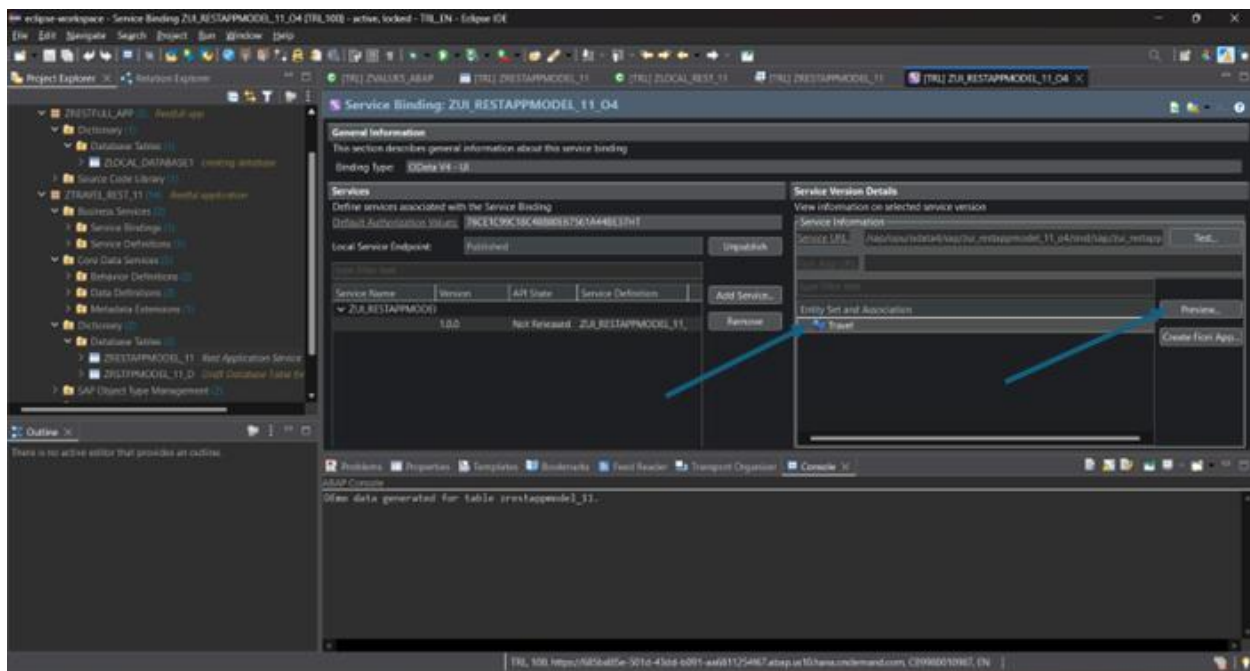
- Business Services is created
- Core Data Services is created
- Click on Publish



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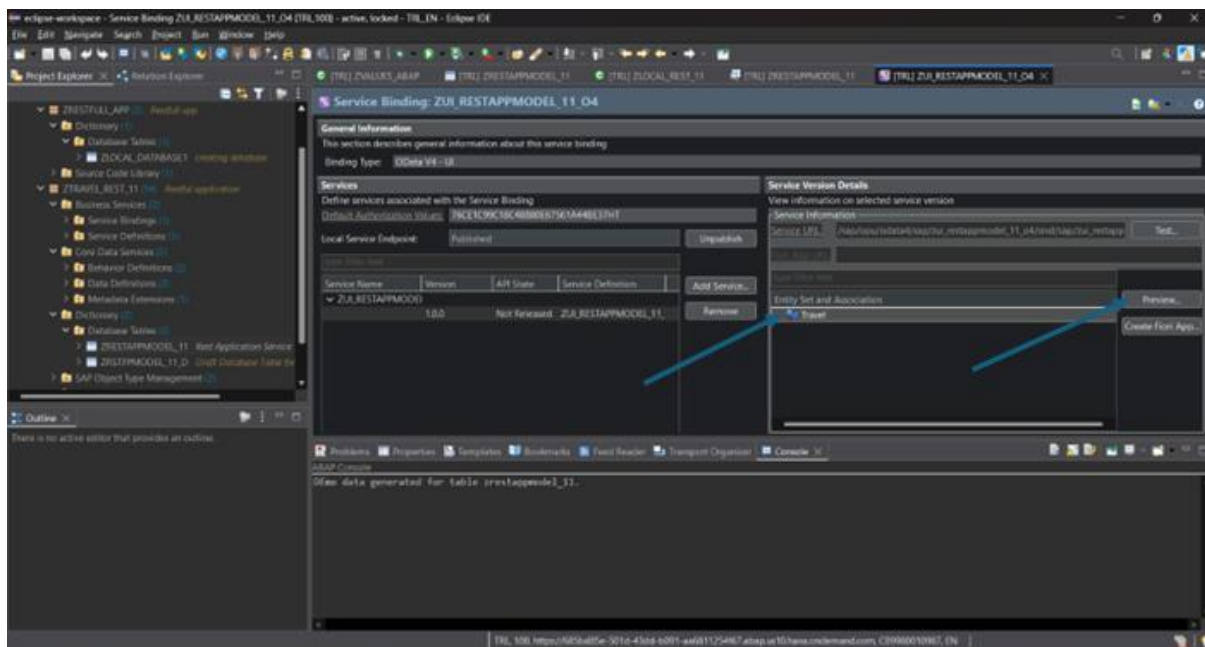
Step 24:- Ones it published.

- Click on the “Travel” and Click on the “Preview”



Step 25:- If you are getting any error

- Click on “Close”
- Again click on the “Travel” and Click on the “Preview”



Step 26:- Ones you click on the “Preview” you can see the “Preview for Fiori Elements App”

- If you want to check the data present in the table, Click on the “Go”

Step 27:- You can “Create” and “Delete” the data in the preview.

Travel ID	Agency ID	Customer ID	Starting Date	End Date	Booking Fee	Total Price	Description	Overall Sta...
1	70041	594	Apr 12, 2025	Feb 8, 2026	40.00	USD 1,792.00	USD Business	A
2	70007	608	Apr 12, 2025	Apr 12, 2025	20.00	USD 896.00	USD Business Trip for Christine, Pierre	A
3	70046	93	Apr 12, 2025	Feb 8, 2026	85.00	USD 3,559.00	USD Vacation	O
4	70042	665	Apr 12, 2025	Feb 8, 2026	40.00	USD 1,792.00	USD Vacation 222	O
5	70007	225	Apr 12, 2025	Apr 12, 2025	20.00	USD 992.00	USD Business Trip for Kurt, Ida	O
6	70049	72	Apr 12, 2025	Feb 8, 2026	120.00	USD 5,586.00	USD Vacation	O
7	70046	138	Apr 12, 2025	Feb 8, 2026	120.00	USD 5,691.00	USD Vacation	O
8	70017	705	Apr 12, 2025	Apr 14, 2025	60.00	USD 2,777.00	USD Vacation for Lother, Lisa, Thilo	O

Step 28:- To check the status of any element, Click on the element or data

- To update the values, Click on the “Edit”

SAP ###GENERATED Core Data Service Entity

1

1

General Information

Travel ID: 1	Booking Fee: 40.00 USD	Mime Type: -	Changed On: Jul 1, 2025, 11:55:22AM
Agency ID: 70041	Total Price: 1,792.00 USD	Filename: -	Changed On: Jul 1, 2025, 11:55:22AM
Customer ID: 594	Description: Business	Created By: Pan	
Starting Date: Apr 12, 2025	Overall Status: A	Created On: Mar 11, 2025, 9:24:50AM	
End Date: Feb 8, 2026	Attachment: -	Changed By: CB9980008398	

Edit **Delete** **Copy**

Step 29:- How to add a record to the database table.

- Click on the “Create”

SAP Preview for Fiori Elements App

Standard

Editing Status: All

Travel ID: Agency ID: Customer ID: Starting Date: End Date:

Booking Fee: Total Price: Description: Overall Status: Mime Type: Filename:

Created By: Created On: Changed By: Changed On: Changed On:

Create **Delete** **Copy** **Filter** **Settings** **Help**

Travel ID	Agency ID	Customer ID	Starting Date	End Date	Booking Fee	Total Price	Description	Overall Sta...
					0.00	0.00	ravi travel	X
1	70041	594	Apr 12, 2025	Feb 8, 2026	40.00	USD 1,792.00	USD Business	A
2	70007	608	Apr 12, 2025	Apr 12, 2025	20.00	USD 896.00	USD Business Trip for Christine, Pierre	A
3	70046	93	Apr 12, 2025	Feb 8, 2026	85.00	USD 3,559.00	USD Vacation	O
4	70042	665	Apr 12, 2025	Feb 8, 2026	40.00	USD 1,792.00	USD Vacation 222	O
5	70007	225	Apr 12, 2025	Apr 12, 2025	20.00	USD 992.00	USD Business Trip for Kurt, Ida	O
6	70049	72	Apr 12, 2025	Feb 8, 2026	120.00	USD 5,586.00	USD Vacation	O
7	70046	138	Apr 12, 2025	Feb 8, 2026	120.00	USD 5,691.00	USD Vacation	O
8	70012	705	Apr 12, 2025	Apr 14, 2025	60.00	USD 2,777.00	USD Vacation for Father Hills, Thilo	O

Step 30:- Enter the sample id “Travel id = 10” and Click on Continue.

The screenshot shows the SAP Fiori Elements App interface. A 'New Object' dialog box is displayed in the center. The 'Travel ID' field is highlighted with a blue border and contains the value '10'. Two black arrows point to the 'Travel ID' field and the 'Continue' button. The background is a blurred view of the app's main screen.

Step 31:- Enter the details

- Agency ID: - 70101
- Customer ID:- 576
- Starting Date: - Apr 16,2025
- Ending Date: Aug 18, 2026
- Booking Fee: - 87.00 USD
- Total Price: - 1175.00 USD
- Overall status: - C

The screenshot shows the SAP Fiori Elements App interface. The 'General Information' form is displayed. The form contains the following fields and values:

Field	Value
Travel ID	10
Agency ID	70101
Customer ID	576
Starting Date	Apr 16, 2025
End Date	Aug 18, 2026
Booking Fee	87.00 USD
Total Price	1,175.00 USD
Description	Business
Overall Status	C
Attachment	
Name Type	
Filename	
Created By	CB9980010987
Created On	Jul 7, 2025, 12:59:07 PM
Changed By	CB9980010987
Changed On	Jul 7, 2025, 1:09:32 PM

Arrows point to the 'Travel ID', 'Agency ID', 'Customer ID', 'Starting Date', 'End Date', 'Booking Fee', 'Total Price', 'Description', 'Overall Status', 'Attachment', 'Name Type', 'Filename', 'Created By', 'Created On', 'Changed By', 'Changed On', and the 'Create' button.

Step 32:- Ones you Click on the Create the sample data is add into database.

###GENERATED Core Data Service Entity

10

Edit Delete

General Information

Travel ID: 10	Booking Fee: 87.00 USD	Mime Type: -	Changed On: Jul 7, 2025, 1:11:38 PM
Agency ID: 70101	Total Price: 1,175.00 USD	Filename: -	Changed On: Jul 7, 2025, 1:11:38 PM
Customer ID: 576	Description: Business	Created By: CB9980010987	
Starting Date: Apr 16, 2025	Overall Status: C	Created On: Jul 7, 2025, 1:11:38 PM	
End Date: Aug 18, 2026	Attachment: -	Changed By: CB9980010987	

Check the last row.

CLUE	TRAVEL	AGENCY	CUSTOMER	BEGIN_DA	END_DA	BOOKING_FEE	TOTAL_PRICE	CURRENCY_CO	DESCRIPTION
100	00000000	000000	000000	2025-04-12	2026-02-08	0.00	0.00	USD	rail travel
100	00000001	070041	000594	2025-04-12	2026-02-08	40.00	1792.00	USD	Business
100	00000002	070007	000608	2025-04-12	2026-02-08	20.00	886.00	USD	Business Trip for C
100	00000003	070046	000093	2025-04-12	2026-02-08	85.00	3559.00	USD	Vacation
100	00000004	070042	000665	2025-04-12	2026-02-08	40.00	1792.00	USD	Vacation 222
100	00000005	070007	000225	2025-04-12	2026-02-08	20.00	992.00	USD	Business Trip for B
100	00000006	070049	000072	2025-04-12	2026-02-08	120.00	5586.00	USD	Vacation
100	00000007	070046	000138	2025-04-12	2026-02-08	120.00	5691.00	USD	Vacation
100	00000008	070012	000705	2025-04-12	2026-02-14	60.00	2777.00	USD	Vacation for Loth
100	00000009	070032	000115	2025-04-12	2026-02-08	120.00	5792.00	USD	Visiting Anna
100	00000010	070101	000576	2025-04-16	2026-08-18	87.00	1175.00	USD	Business